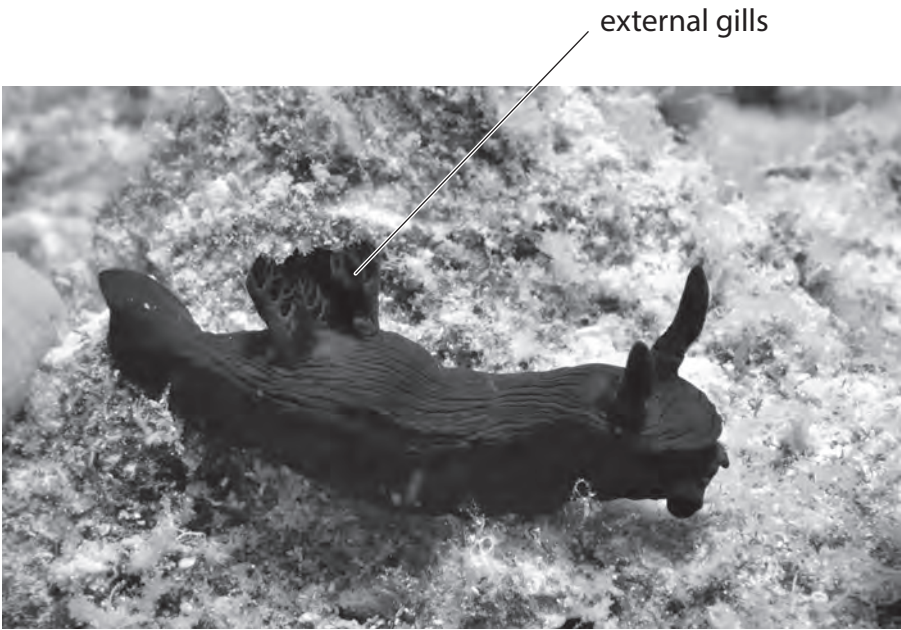


3 The photograph shows a sea slug of the genus *Aplysia*, a marine animal.



(Source: © Walkdragon / Shutterstock)

Aplysia can be kept in tanks in a laboratory.

Aplysia absorbs oxygen using external gills.

When the gills are touched they are withdrawn into the body by a reflex action.

A student applied a touch stimulus to the gills and recorded the percentage of the gills that remained visible.

The touch stimulus was repeated five times at 30 second intervals and each response recorded.

The investigation was repeated with two more sea slugs and the means calculated.

The table shows the results obtained from these three sea slugs.

Stimulus	1st	2nd	3rd	4th	5th
Mean percentage of gill that remained visible (%)	10	52	47	49	45

(a) State a suitable null hypothesis for this investigation.

(1)

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(b) Draw a suitable graph to show the results in this table.

(3)



(c) The student used a correlation coefficient (r_s) to analyse the results.

The student produced a table to calculate the correlation coefficient (r_s).

Stimulus number (a)	Mean percentage of visible gills (b)	Rank (a)	Rank (b)	d	d^2
1	10	1	1	0	
2	52	2	5	-3	
3	47	3	3	0	
4	49	4	4	0	
5	45	5	2	-3	

(i) Complete the table.

(1)

(ii) Calculate the correlation coefficient, r_s , using the formula:

(2)

$$r_s = 1 - \frac{6(\sum d^2)}{n(n^2 - 1)}$$

Where:

$\sum$ = the sum of

d = the difference between each pair of ranks

n = the size of the sample (number of pairs of values)

Answer

(iii) The table shows some critical values for this statistical test.

Number of pairs of values	Level of significance (p)		
	0.10	0.05	0.01
4	1.000	–	–
5	0.900	1.000	–
6	0.829	0.886	1.000
7	0.714	0.786	0.929
8	0.643	0.738	0.881
9	0.600	0.700	0.833
10	0.564	0.648	0.794

Describe the conclusion that can be drawn from this investigation.

Use your calculated r_s value and the table of critical values to support your answer.

(2)

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(d) Describe how the student could extend this investigation to collect more data to either support or reject the null hypothesis, in part (a).

(2)

(e) The change in response of an animal to repeated stimulation is an example of habituation.

Give **one** reason why habituation is beneficial to an animal.

(1)

(Total for Question 3 = 12 marks)